ARML Competition 2026

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1 Team Round

Problem 1. Given that the product 20262025×20262023 is the fifteen-digit number $410549\underline{A}165765\underline{B}5$, compute the ordered pair (A, B).

Problem 2. Consider the following cross-number puzzle. No answer has a leftmost digit of 0.

1-Across: Sum of the cubes of the digits in 4-Across
4-Across: A perfect cube
5-Across: Prime with a digit sum of 16

1-Down: Its digits form an increasing arithmetic sequence
2-Down: A Fibonacci number
3-Down: Decimal representation of 1101000101_2

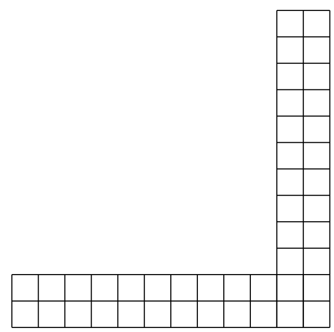
$^1 A$	$^2 B$	$^3 C$
$^4 D$	E	F
$^5 G$	H	I

Compute $\underline{A}\,\underline{B}\,\underline{C} + \underline{D}\,\underline{E}\,\underline{F} + \underline{G}\,\underline{H}\,\underline{I}$.

Problem 3. Let $RAIS$ be a square with rational side length. Point T lies on side $\overline{AI}$ so that $\frac{[STAR]}{[SIT]} = 10$. Given that TS^2 is an integer, compute the least possible value of TS^2.

Problem 4. Let P be the infinite product $\sqrt[3]{2} \times \sqrt[9]{4} \times \sqrt[27]{8} \times \sqrt[81]{16} \times \cdots \times \sqrt[3^n]{2^n} \times \cdots$. Compute P^4.

Problem 5. Compute the number of ways to tile the grid shown at right with 2×1 and 1×2 rectangles.



Problem 6. Compute the least real number c such that the graphs of $y = x^2 + c$ and $x = y^2 + c$ intersect at an odd number of points.

Problem 7. An *Alphabubu Blind Box* is a box containing one tile, on which is one letter of the alphabet. When Glenn opens a box, the tile inside is equally likely to contain each of the 26 letters of the alphabet. Compute the expected number of boxes that Glenn would need to open in order to spell GLENN with his tiles.

Problem 8. In rectangle $ARML$, point X lies on side $\overline{AR}$, and point Y lies on side $\overline{RM}$. The line that passes through Y and is perpendicular to $\overleftrightarrow{MX}$ intersects $\overleftrightarrow{MX}$ and $\overleftrightarrow{MA}$ at P and Q, respectively. Given that $AX = 14$, $XR = 10$, and $7 \cdot YP = 13 \cdot PQ$, compute $[ARML]$.

Problem 9. Let a and b be positive real numbers such that $\dfrac{a+b}{2}$, $\sqrt{ab}$, and $\dfrac{2}{\frac{1}{a} + \frac{1}{b}}$ are integers with sum 335. Compute $|a - b|$.

Problem 10. Let $\triangle ABC$ be a nondegenerate acute triangle. A student computes the six standard trigonometric functions (i.e., sin, cos, tan, csc, sec, and cot) of each of the three angles of $\triangle ABC$, resulting in k distinct values. Compute the sum of all possible values of k.

2 Team Round Answers

Answer 1. $(6, 7)$

Answer 2. 1008

Answer 3. 5

Answer 4. 8

Answer 5. 25632

Answer 6. $\frac{1}{4}$ (or 0.25)

Answer 7. $\frac{1625}{24}$ $\left(\text{or } 67\frac{17}{24}\right)$

Answer 8. $120\sqrt{6}$

Answer 9. $210\sqrt{5}$

Answer 10. 113

3 Team Round Solutions

Problem 1. Given that the product 20262025×20262023 is the fifteen-digit number $410549\underline{A}165765\underline{B}5$, compute the ordered pair (A, B).

Solution 1. Observing that 20262024 is a multiple of 3, express this product generally as $(3n+1)(3n-1) = 9n^2 - 1$ for some value of n. This means that $410549\underline{A}165765\underline{B}6$ is a multiple of 9, and so the sum of its digits must be divisible by 9. Thus

$$4 + 1 + 5 + 4 + 9 + A + 1 + 6 + 5 + 7 + 6 + 5 + B + 6 + 1 = 59 + A + B$$

must be a multiple of 9. Quick inspection of the product $25 \cdot 23 = 575$ implies that B must be equal to 7. Thus $66 + A$ must be 72, so $A = 6$, and the answer is $\mathbf{(6, 7)}$.

Problem 2. Consider the following cross-number puzzle. No answer has a leftmost digit of 0.

1-Across: Sum of the cubes of the digits in 4-Across
4-Across: A perfect cube
5-Across: Prime with a digit sum of 16

1-Down: Its digits form an increasing arithmetic sequence
2-Down: A Fibonacci number
3-Down: Decimal representation of 1101000101_2

[1] A	[2] B	[3] C
[4] D	E	F
[5] G	H	I

Compute $\underline{A}\,\underline{B}\,\underline{C} + \underline{D}\,\underline{E}\,\underline{F} + \underline{G}\,\underline{H}\,\underline{I}$.

Solution 2. Although one can convert the number in 3-Down (call it N) to decimal, one could also estimate

$$N > 2^9 + 2^8 > 700_{10},$$

so as a decimal, the leftmost digit of N (which is C) must be at least 7.

Because 2-Down and 4-Across share the same middle digit, it is useful to list three-digit Fibonacci numbers and perfect cubes. Note that 144 is a Fibonacci number and shares its middle digit with 343, a perfect cube. Note also that 610 is a Fibonacci number and shares its middle digit with the perfect cubes 216 and 512. However, the sums of the cubes of the digits of 216 and 512 (as required by 1-Across) are 225 and 134, respectively, and in both cases, the units digits are less than 7.

Thus 2-Down is 144, 4-Across is 343, and 1-Across is $3^3 + 4^3 + 3^3 = 118$. The clue for 1-Down implies that $G = 5$. Because 5-Across has a digit sum of 16, it follows that $I = 16 - (5 + 4) = 7$, and one can verify that 547 is indeed prime. The value of I can also be determined by adding up units digits of appropriate powers of 2. Thus $\underline{A}\,\underline{B}\,\underline{C} + \underline{D}\,\underline{E}\,\underline{F} + \underline{G}\,\underline{H}\,\underline{I} = 118 + 343 + 547 = \mathbf{1008}$. The completed grid is shown below.

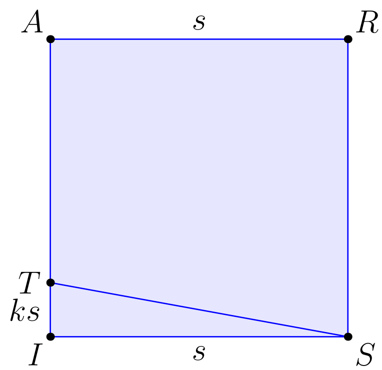
1	1	8
3	4	3
5	4	7

Problem 3. Let $RAIS$ be a square with rational side length. Point T lies on side $\overline{AI}$ so that $\frac{[STAR]}{[SIT]} = 10$. Given that TS^2 is an integer, compute the least possible value of TS^2.

Solution 3. Let the side length of $RAIS$ be s, and let $TI = ks$. Then $[SIT] = \frac{1}{2}ks^2$ and $[STAR] = (1 - \frac{1}{2}k)s^2$, so

$$\frac{1 - \frac{1}{2}k}{\frac{1}{2}k} = 10,$$

so $k = \frac{2}{11}$. Thus $TS^2 = \frac{125}{121}s^2$.



If TS^2 is an integer, then it must be divisible by 5. Indeed, the exponent of 5 in the (rational) prime factorization of TS^2 must be odd and hence at least 1. Note that $TS^2 = 5$ is achievable by taking $s = \frac{11}{5}$. So the least possible integral value of TS^2 is **5**.

Problem 4. Let P be the infinite product $\sqrt[3]{2} \times \sqrt[9]{4} \times \sqrt[27]{8} \times \sqrt[81]{16} \times \cdots \times \sqrt[3^n]{2^n} \times \cdots$. Compute P^4.

Solution 4. Note that $\log_2 P = \frac{1}{3} + \frac{2}{9} + \frac{3}{27} + \frac{4}{81} + \cdots$ and $\frac{1}{3}\log_2 P = \frac{0}{3} + \frac{1}{9} + \frac{2}{27} + \frac{3}{81} + \cdots$. Subtracting, $\frac{2}{3}\log_2 P = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \cdots = \frac{1}{2}$. Thus $\log_2 P = \frac{3}{4}$ so $P^4 = \mathbf{8}$.

Alternate Solution: Note that $P = 2^S$, where

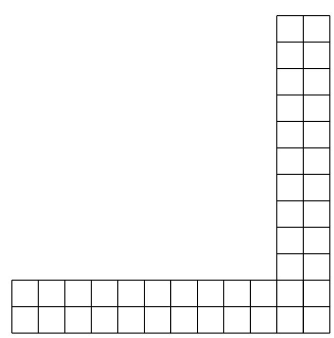
$$S = \sum_{k=1}^{\infty} \frac{k}{3^k}.$$

Writing out S term by term and multiplying the resulting equation by $\frac{1}{3}$ yields

$$\begin{aligned} S &= \frac{1}{3} + \frac{2}{9} + \frac{3}{27} + \frac{4}{81} + \cdots \\ \frac{1}{3}S &= \frac{1}{9} + \frac{2}{27} + \frac{3}{81} + \frac{4}{243} + \cdots. \end{aligned}$$

Taking the differences of both sides of the above equation yields $\frac{2}{3}S = \frac{1}{3^1} + \frac{1}{3^2} + \frac{1}{3^3} + \frac{1}{3^4} + \cdots = \frac{1/3}{1-1/3} = \frac{1}{2}$, hence $S = \frac{3}{4}$, and $P^4 = (2^{3/4})^4 = 2^3 = \mathbf{8}$.

Problem 5. Compute the number of ways to tile the grid shown at right with 2×1 and 1×2 rectangles.



Solution 5. It can be readily shown that the number of ways to tile a $2 \times N$ rectangle with 2×1 rectangles is F_{N+1}, the $(N+1)^{\text{st}}$ Fibonacci number (with $F_0 = 0$ and $F_1 = 1$).

Accordingly, the problem can be broken down by how the four squares at the intersection of the horizontal and vertical 2×12 parts are tiled. If the four squares are tiled by two horizontal or vertical tiles, then there are $(F_{11})^2$ ways to tile the remaining 2×10 and 10×2 rectangles. If the squares are covered by one horizontal tile against the bottom boundary and two vertical tiles covering the other two squares (or similarly, one vertical tile against the rightmost boundary and two horizontal tiles covering the other two squares), then there are $F_{11} \cdot F_{10}$ ways to tile the remaining rectangles. Because $F_{10} = 55$ and $F_{11} = 89$, the answer is $2 \cdot 89^2 + 2 \cdot 89 \cdot 55 = \mathbf{25632}$.

Problem 6. Compute the least real number c such that the graphs of $y = x^2 + c$ and $x = y^2 + c$ intersect at an odd number of points.

Solution 6. For most $c \geq 0$, there are 2 intersection points for small c and none for large c. Because the two parabolas are reflections over the line $y = x$, the transition happens when the two parabolas intersect along the axis of reflection, so $x = x^2 + c$ has a unique solution, or $x^2 - x + c$ is a perfect square, which happens when $c = 1/4$.

For $c < 0$, the number of intersections goes from 2 to 4, but without a transition of 3 intersection points. Instead there are two intersections when $c \geq -3/4$ and four intersections when $c < -3/4$.

One way to see this is that intersections happen when $x = (x^2 + c)^2 + c$, or $0 = x^4 + 2cx^2 - x + c^2 + c = (x^2 - x + c)(x^2 + x + 1 + c)$. The first factor has a double root when $c = 1/4$, as already found, and the second has a double root when the discriminant $1 - 4 - 4c = 0$, or $c = -3/4$. But when $c = -3/4$, the first factor $x^2 - x - 3/4$ is 0 when $x = -1/2$ or $3/2$, while the second factor $x^2 + x + 1/4$ has its double root at $x = -1/2$, so the degree-four polynomial has a triple root at $-1/2$ and single root at $3/2$ thus the parabolas intersect twice. Therefore an odd number of intersections only happens when c is $\mathbf{\frac{1}{4}}$.

Problem 7. An *Alphabubu Blind Box* is a box containing one tile, on which is one letter of the alphabet. When Glenn opens a box, the tile inside is equally likely to contain each of the 26 letters of the alphabet. Compute the expected number of boxes that Glenn would need to open in order to spell GLENN with his tiles.

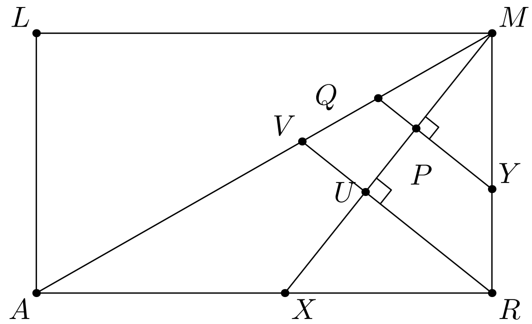
Solution 7. The expected number of boxes to open is the reciprocal of the probability of opening those tiles. For example, there are four letter tiles needed, the expected number of boxes to open any one of the G, L, E, or N is $\frac{26}{4} = 6.5$ boxes, so in order to spell GLEN, the total expected number of boxes needed to open is $\frac{26}{4} + \frac{26}{3} + \frac{26}{2} + \frac{26}{1} = 3256$. Proceed by breaking the problem into four cases, depending on whether or not the first of the two Ns needed is opened first, second, third, or fourth:

- **Case 1:** The first N is opened first. Then $\frac{26}{4} + \frac{26}{4} + \frac{26}{3} + \frac{26}{2} + \frac{26}{1} = \frac{182}{3}$.
- **Case 2:** The first N is opened second. Then $\frac{26}{4} + \frac{26}{3} + \frac{26}{3} + \frac{26}{2} + \frac{26}{1} = \frac{377}{6}$.
- **Case 3:** The first N is opened third. Then $\frac{26}{4} + \frac{26}{3} + \frac{26}{2} + \frac{26}{2} + \frac{26}{1} = \frac{403}{6}$.
- **Case 4:** The first N is opened fourth. Then $\frac{26}{4} + \frac{26}{3} + \frac{26}{2} + \frac{26}{1} + \frac{26}{1} = \frac{481}{6}$.

Summing the expectations of the four cases, each term $\frac{26}{4}, \frac{26}{3}, \frac{26}{2}, \frac{26}{1}$ is added five times, and each of these four cases is equally likely, so compute $\frac{5}{4}(\frac{26}{4} + \frac{26}{3} + \frac{26}{2} + \frac{26}{1}) = \mathbf{\frac{1625}{24}}$, (which is approximately 67.7 boxes).

Problem 8. In rectangle $ARML$, point X lies on side $\overline{AR}$, and point Y lies on side $\overline{RM}$. The line that passes through Y and is perpendicular to $\overleftrightarrow{MX}$ intersects $\overleftrightarrow{MX}$ and $\overleftrightarrow{MA}$ at P and Q, respectively. Given that $AX = 14$, $XR = 10$, and $7 \cdot YP = 13 \cdot PQ$, compute $[ARML]$.

Solution 8. Let the line passing through R and parallel to $\overline{YQ}$ intersect $\overline{MX}$ and $\overline{MA}$ in points U and V, where $7 \cdot RU = 13 \cdot UV$.



Using Menelaus' Theorem for triangle ARV and transversal $\overleftrightarrow{MX}$ yields

$$1 = \frac{AX}{XR} \cdot \frac{RU}{UV} \cdot \frac{MV}{MA} = \frac{14}{10} \cdot \frac{13}{7} \cdot \frac{MV}{MA},$$

hence $\frac{MV}{MA} = \frac{5}{13}$ and $\frac{MV}{VA} = \frac{5}{8}$. Using Menelaus' Theorem for triangle AMX and transversal $\overleftrightarrow{VR}$ yields

$$1 = \frac{AV}{VM} \cdot \frac{MU}{UX} \cdot \frac{XR}{RA} = \frac{8}{5} \cdot \frac{MU}{UX} \cdot \frac{10}{24},$$

hence $\frac{MU}{UX} = \frac{3}{2}$. Thus let $MU = 3x$ and $UX = 2x$. Because $\overline{RU}$ is an altitude in right triangle MRX, from similar triangles XUR and XRM, it follows that $XR^2 = XU \cdot XM$. Therefore $10^2 = 2x \cdot 5x$, so $x = \sqrt{10}$, and thus $MX = 5x = 5\sqrt{10}$, $MR = 5\sqrt{6}$. Because $AR = 24$, it follows that $[ARML] = \mathbf{120\sqrt{6}}$.

Problem 9. Let a and b be positive real numbers such that $\dfrac{a+b}{2}$, $\sqrt{ab}$, and $\dfrac{2}{\frac{1}{a}+\frac{1}{b}}$ are integers with sum 335. Compute $|a-b|$.

Solution 9. Note that $AH = G^2$. Let $n = \gcd(A, H)$, so $\frac{A}{n}$ and $\frac{H}{n}$ are relatively prime positive integers whose product is a perfect square. Hence they must themselves be perfect squares; say p^2 and q^2. Henceforth, write $A = np^2$, $G = npq$, $H = nq^2$ for integers n and $p \geq q > 0$.
Thus

$$n(p^2 + pq + q^2) = 5 \cdot 67.$$

Because $p^2 + pq + q^2 = (p-q)^2 + 3pq$ is 0 mod 3 or 1 mod 3, we must have $p^2 + pq + q^2 = 67$, and $n = 5$. The only possible (p, q) is $(7, 2)$ so $(A, G, H) = (245, 70, 20)$. Thus

$$(a-b)^2 = (a+b)^2 - 4ab = (2A)^2 - 4G^2 = 490^2 - 4 \cdot 70^2 = 70^2(7^2 - 4)$$

and so $|a-b| = 70\sqrt{5 \cdot 9} = \mathbf{210\sqrt{5}}$.

Problem 10. Let $\triangle ABC$ be a nondegenerate acute triangle. A student computes the six standard trigonometric functions (i.e., sin, cos, tan, csc, sec, and cot) of each of the three angles of $\triangle ABC$, resulting in k distinct values. Compute the sum of all possible values of k.

Solution 10. Before delving into casework, here are some observations that help in all cases. Define a new operator $\operatorname{tot}(t) = \min\{\tan t, \cot t\}$. Note that $\operatorname{tot}(t) < 1$ for $0^\circ < t < 90^\circ$ and $t \neq 45^\circ$.

Next, let ϵ be the number of angles A, B, C which are equal to 45°. Note that $\epsilon = 0$ or $\epsilon = 1$. Then k is ϵ less than twice the number of values in the following 3×3 matrix:

$$M = \begin{bmatrix} \sin A & \cos A & \operatorname{tot} A \\ \sin B & \cos B & \operatorname{tot} B \\ \sin C & \cos C & \operatorname{tot} C \end{bmatrix}.$$

Note that within each column, two entries can only be equal only if the corresponding angles are equal. (For the tot column, this requires acuteness of the triangle.) Also, $\sin A = \cos B$ and similar equations are impossible because ABC is an acute triangle; and $\sin A = \cos A$ can only occur if $A = 45°$.

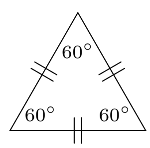
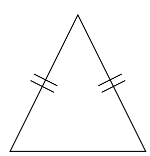
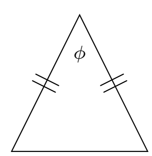
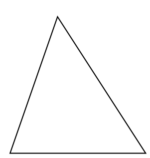
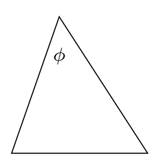
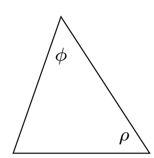
It will also be necessary to analyze the equations $\sin(x) = \operatorname{tot}(x)$ and $\cos(x) = \operatorname{tot}(x)$. For acute angles x, note that $\sin(x) = \operatorname{tot}(x)$ occurs only if $45° < x < 90°$ and $\sin(x) = \cot(x) = \frac{\cos(x)}{\sin(x)}$; this is equivalent to $\cos^2 x + \cos x = 1$, which occurs only when $\cos(x) = \frac{1}{2}(-1 + \sqrt{5})$. In other words, $\sin(x) = \operatorname{tot}(x)$ has a unique solution, henceforth denoted by $\phi = \arccos\left(\frac{1}{2}(-1 + \sqrt{5})\right)$. (For concreteness, it turns out that $\phi \approx 51.827°$.) By a symmetric analysis, $\cos(x) = \operatorname{tot}(x)$ can only occur when $x = 90° - \phi$.

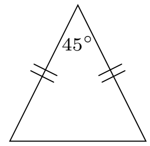
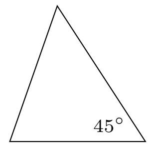
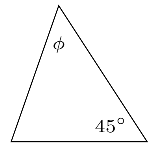
Proceed with case analysis as follows. Call a real number *evil* if it appears more than once in M in different columns, and does not occur because $\sin 45° = \cos 45°$. Then the above properties imply that besides duplicate rows for an isosceles triangle, an evil number must appear exactly twice: once in the tot column and once in a sin or cos column.

Let n be the number of evil numbers, and let r be the number of different rows (that is, $r = 1, 2, 3$ if ABC is equilateral, isosceles and non-equilateral, or scalene, respectively). Note that $n \le r - \epsilon$ because all evil numbers appear once in the last column and lie in the interval $(0, 1)$. Then

$$k = 2(3r - n - \epsilon) - \epsilon = 6r - 2n - 3\epsilon.$$

The table below shows the possible cases in consideration, together with a triangle achieving each possible configuration. In the cases where an angle is underspecified, the choice should be made "sufficiently generically", so that no unexpected equalities arise.

$\epsilon = 0$	$n = 0$	$n = 1$	$n = 2$	$n = 3$
Equilateral $r = 1$	 $k = 6$			
Isosceles $r = 2$	 $k = 12$	 $k = 10$	**Impossible** $k = 8$	
Scalene $r = 3$	 $k = 18$	 $k = 16$	 $k = 14$	**Impossible** $k = 12$

$\epsilon = 1$	$n = 0$	$n = 1$	$n = 2$	$n = 3$
Isosceles $r = 2$	 $k = 9$	**Impossible** $k = 7$		
Scalene $r = 3$	 $k = 15$	 $k = 13$	**Impossible** $k = 11$	

In the exceptional case where $(\epsilon, n, r) = (0, 2, 3)$, the angle ϕ is defined so that $45° < \phi < 90°$ and the equation

$$\sin\bigl(180° - (\phi + \rho)\bigr) = \cot \rho$$

holds. This is possible because if ρ were equal to $45°$, the right-hand side of the above equation would be greater while when $\rho = 90°$, the left-hand side would be greater, so continuity guarantees the existence of one such ρ. (For concreteness, it turns out that $\rho \approx 45.217°$.)

It remains to show that the cases marked "Impossible" indeed never occur. The basic heuristic argument is that in each of these cases, one could consider writing down a system of equations in terms of the angles. In all these cases, there is one more equation than variable, so the resulting equation should never have any solutions. Thus one can rule out these cases based solely on intuition; hence the sum of the possible values of k is

$$6 + 12 + 10 + 18 + 16 + 14 + 9 + 15 + 13 = \mathbf{113}.$$

For completeness, a proof of the impossible cases is sketched below. In order to make the story more readable, machine-based numerical approximations are used to describe the solutions to one-variable equations, even though this does not constitute a rigorous proof. To write an airtight formal proof, the numerical approximations should be replaced by working with the values of $\sin\theta$ and $\cos\theta$ directly, showing that the pairs of equations cannot hold simultaneously. The organizers of the competition stress that it is not expected that contestants would carry out these proofs.

- The case where $\epsilon = 1$ and $r = 2$ is easy because any isosceles acute triangle with an angle equal to $45°$ must in fact have the other two angles equal to $67.5°$, i.e., there is only one triangle up to similarity, and it satisfies $n = 0$.
- Consider the case $(\epsilon, n, r) = (0, 2, 2)$. If any of the angles are equal to ϕ or $90° - \phi$, then $n = 1$. Otherwise, let $A = 2\theta$, $B = C = 90° - \theta$, where $0° < \theta < 45°$. The angle θ would have to obey

$$\begin{aligned} \operatorname{tot}(90° - \theta) &\in \{\sin 2\theta, \cos 2\theta\} \\ \operatorname{tot}(2\theta) &\in \{\sin\theta, \cos\theta\}. \end{aligned}$$

  However, the unique angle $0 < \theta < 45°$ for which the first equation holds is $\theta \approx 28.532°$ for which $\tan\theta = \cos(2\theta)$; this does not satisfy the second equation.
- Consider the case $(\epsilon, n, r) = (1, 2, 3)$. If any of the angles are equal to ϕ or $90° - \phi$, then $n = 1$. Otherwise, let $A = \theta$ and $B = 135° - \theta$, where $45° < \theta < 90°$. In order for $n \geq 2$ to occur, the angle θ would have to obey

$$\begin{aligned} \operatorname{tot}(\theta) &\in \left\{\sin(135° - \theta), \cos(135° - \theta), \frac{1}{\sqrt{2}}\right\} \\ \operatorname{tot}(135° - \theta) &\in \left\{\sin\theta, \cos\theta, \frac{1}{\sqrt{2}}\right\}. \end{aligned}$$

  However, the only angles $45° < \theta < 90°$ satisfying the first equation are $\theta = \arctan(\sqrt{2}) \approx 54.736°$ and $\theta \approx 68.366°$, for which $\cot\theta = \cos(135° - \theta)$. This does not satisfy the second equation.

- The case $(\epsilon, n, r) = (0, 3, 3)$ is the most difficult because the family of scalene triangles is parametrized by two variables. Therefore, even given numerical analysis, this case cannot be easily dispatched. However, handling this is also not necessary for the problem at hand! This case yields the value $k = 12$, which was already covered by $(\epsilon, n, r) = (0, 0, 2)$. Therefore, while this case does turn out to be impossible, proving the impossibility is irrelevant to the final answer.

4 Power Round 2026: Abundance and (Im)perfection

Instructions: The power question is worth 50 points; each part's point value is given in brackets next to the part. To receive full credit, the presentation must be legible, orderly, clear, and concise. If a problem says "list" or "compute," you need not justify your answer. If a problem says "determine," "find," or "show," then you must show your work or explain your reasoning to receive full credit, although such explanations do not have to be lengthy. If a problem says "justify" or "prove," then you must prove your answer rigorously. Even if not proved, earlier numbered items may be used in solutions to later numbered items, but not vice versa. Pages submitted for credit should be NUMBERED IN CONSECUTIVE ORDER AT THE TOP OF EACH PAGE in what your team considers to be proper sequential order. PLEASE WRITE ON ONLY ONE SIDE OF THE ANSWER PAPERS. Put the TEAM NUMBER (not the team name) on the cover sheet used as the first page of the papers submitted. Do not identify the team in any other way.

For a positive integer n, let $\sigma(n)$ denote the sum of the positive divisors of n. For example, $\sigma(3) = 1 + 3 = 4$, $\sigma(4) = 1 + 2 + 4 = 7$, and $\sigma(6) = 1 + 2 + 3 + 6 = 12$. Note that because 1 and n are both divisors, $\sigma(n) > n$ for all $n > 1$.

Define the *abundancy* of n to be $A(n) = \dfrac{\sigma(n)}{n}$. For example, $A(3) = \dfrac{4}{3}$ and $A(4) = \dfrac{7}{4}$.

In the problems below, you may use the following facts without proof:

- **Fact 1:** If a and b are relatively prime positive integers, then $\sigma(ab) = \sigma(a)\sigma(b)$.
- **Fact 2:** The sum of the reciprocals of the prime numbers, $\displaystyle\sum_{j=1}^{\infty} \frac{1}{p_j}$, is infinite (i.e., grows without bound).

1. Compute $A(7)$, $A(8)$, and $A(9)$. [3 pts]

In what follows, the letters p and q will always denote distinct primes and all other letters will denote positive integers unless otherwise stated.

2. **a.** Find a closed-form expression for $A(p^k)$. Your expression must not contain summations or "…". [2 pts]

b. Find the least upper bound for $A(p^k)$, where p is fixed and k can vary. [2 pts]

c. Show that if a and b are relatively prime positive integers, then $A(ab) = A(a)A(b)$. [1 pt]

d. Find the least upper bound for $A(p^k q^m)$, where p and q are fixed and k and m can vary. [2 pts]

e. Given primes $p_1, p_2, \ldots, p_k$, compute the least upper bound of the set of $A(n)$, where n ranges over integers whose prime factors are all in the set $\{p_1, p_2, \ldots, p_k\}$. [1 pt]

Note that $A(6) = \dfrac{1+2+3+6}{6} = \dfrac{12}{6} = 2$. Numbers with $A(n) = 2$ are called *perfect numbers* and have been studied for thousands of years… so that's *not* what this Power Question is about….

3. Prove that if m divides n, then $A(m) \le A(n)$. [4 pts]

4. **a.** Prove that if $n = 2^k 3^m$ for positive integers k and m, then $2 \le A(n) < 3$. [2 pts]

b. Prove that if $A(n) = 3$, then n must have at least three distinct prime factors. [2 pts]

c. Find a number n such that $A(n) = 3$. [2 pts]

d. Find the minimum number of distinct prime factors an integer n can have, given that $A(n) = 4$. Justify your answer. [4 pts]

5. Find the minimum number of distinct prime factors an integer n can have, given that $A(n) \ge 5$. Justify your answer. [3 pts]

6. Prove that $A(n)$ is unbounded; that is, prove that there is no real number C such that $A(n) \le C$ for all n. [3 pts]

A rational number $x > 1$ is called an *abundancy outlaw* if there is no integer n such that $A(n) = x$.

7. Prove that if $m < k < \sigma(m)$ and k is relatively prime to m, then $\dfrac{k}{m}$ is an abundancy outlaw. [3 pts]

8. Find two examples of abundancy outlaws. [2 pts]

9. Prove that for any n that is an odd perfect number, $A(2n) = 3$. [2 pts]

10. Suppose a number n exists such that $A(n) = \dfrac{5}{3}$.

- **a.** Prove that n must be odd. [2 pts]
- **b.** Prove that n must be a multiple of 9. [2 pts]
- **c.** Prove that n must **not** be divisible by 5. [2 pts]
- **d.** Prove that $5n$ must be an odd perfect number. [2 pts]

11. Find a positive integer n such that $A(n) = 5$. You may express your answer as a product of powers of primes. [4 pts]

5 Power Round Solutions

1. **a.** $A(7) = \dfrac{1+7}{7} = \dfrac{8}{7}$

   b. $A(8) = \dfrac{1+2+4+8}{8} = \dfrac{15}{8}$

   c. $A(9) = \dfrac{1+3+9}{9} = \dfrac{13}{9}$

2. **a.** By definition and summing a geometric series, $A(p^k) = \dfrac{1+p+p^2+\cdots+p^k}{p^k} = \dfrac{p^{k+1}-1}{p^k(p-1)}$.

   b. By 2(a), $A(p^k) = \dfrac{p^{k+1}-1}{p^{k+1}-p^k} = \dfrac{p-p^{-k}}{p-1}$. As k increases, p^{-k} decreases toward 0, so the least upper bound is $\dfrac{p}{p-1}$.

   c. Using Fact 1 for the second equality, $A(ab) = \dfrac{\sigma(ab)}{ab} = \dfrac{\sigma(a)\sigma(b)}{ab} = \dfrac{\sigma(a)}{a} \cdot \dfrac{\sigma(b)}{b}$, which is $A(a)A(b)$, by definition.

   d. By part 2(c), $A(p^k q^m) = A(p^k)A(q^m)$. By part 2(b), the least upper bounds for $A(p^k)$ and $A(q^m)$ are $\dfrac{p}{p-1}$ and $\dfrac{q}{q-1}$, respectively. It follows that the least upper bound for $A(p^k)A(q^m)$ is $\dfrac{p}{p-1} \cdot \dfrac{q}{q-1}$.

   e. The abundancy of a product of powers of primes $p_1, p_2, \ldots, p_k$ has a least upper bound of $\displaystyle\prod_{j=1}^{k} \frac{p_j}{p_j - 1}$.

3. First, prove that if p is prime and $k_1 < k_2$, then $A(p^{k_1}) < A(p^{k_2})$. To that end, note that $A(p^k) = \dfrac{1+p+p^2+\cdots+p^k}{p^k} = 1 + p^{-1} + p^{-2} + \cdots + p^{-k}$. Thus if $k_1 < k_2$, then $A(p^{k_2}) - A(p^{k_1}) = p^{-(k_1+1)} + p^{-(k_1+2)} + \cdots + p^{-k_2} > 0$. Note that this also holds if $k_1 = 0$, since $A(1) = 1$.

   Now suppose m divides n. Let the prime factorization of n be $\displaystyle\prod_{j=1}^{r} p_j^{k_j}$. Then $m = \displaystyle\prod_{j=1}^{r} p_j^{m_j}$, where $m_j \leq k_j$ for each j (and m_j may be zero). From the above conclusion, $A(p_j^{m_j}) \leq A(p_j^{k_j})$ for each j. Also, by Problem 2(c),

   $$A(n) = \prod_{j=1}^{r} A(p_j^{k_j}) \quad \text{and} \quad A(m) = \prod_{j=1}^{r} A(p_j^{m_j}).$$

   It then follows that $A(m) \leq A(n)$, because each term in the product for $A(m)$ is less than or equal to the corresponding term in the product for $A(n)$.
   Alternate Solution: Similar to the calculation for p^k in the above solution, $A(n)$ is the sum of the inverses of the divisors of n. When m divides n, the divisors of n include all of the divisors of m, thus $A(m) \leq A(n)$ with equality if and only if $m = n$.

4. **a.** First note that $A(6) = 2$, and 6 divides $2^k 3^m$, thus by Problem 3, $2 \leq A(2^k 3^m)$. By the answer to Problem 2(d) (or 2(e)), $A(2^k 3^m) \leq \frac{2}{1} \cdot \frac{3}{2} = 3$, and the inequality is strict because k and m are finite.

   b. The least upper bound $\frac{p}{p-1} = \frac{1}{1-p^{-1}}$ decreases for greater primes p, thus the maximum possible abundancy for a product of a certain number of distinct prime divisors is achieved with the least possible primes. Combining with Problem 4a, a number with only two prime factors has abundancy less than 3, therefore n must have at least three distinct prime factors.

   c. The known examples are $120 = 2^3 \cdot 3 \cdot 5$, $672 = 2^5 \cdot 3 \cdot 7$, and the much less likely answers of $2^9 \cdot 3 \cdot 11 \cdot 31$, $2^8 \cdot 5 \cdot 7 \cdot 19 \cdot 37 \cdot 73$, $2^{13} \cdot 3 \cdot 11 \cdot 43 \cdot 127$, and $2^{14} \cdot 5 \cdot 7 \cdot 19 \cdot 31 \cdot 151$.

   d. The answer is 4, as demonstrated by $2^5 \cdot 3^3 \cdot 5 \cdot 7$, which works because $A(2^5 \cdot 3^3 \cdot 5 \cdot 7) = \frac{63}{32} \cdot \frac{40}{27} \cdot \frac{6}{5} \cdot \frac{8}{7} = 4$. Three primes cannot work because following the method of Problem 4(b), the three least primes are 2, 3, and 5 so the upper bound for $A(n)$ would be $2 \cdot \frac{3}{2} \cdot \frac{5}{4} = \frac{15}{4} < 4$.

5. Five primes are not enough because the upper bound from Problem 2(e) is $2 \cdot \frac{3}{2} \cdot \frac{5}{4} \cdot \frac{7}{6} \cdot \frac{11}{10} = \frac{77}{16} < 5$, but with six primes, $2 \cdot \frac{3}{2} \cdot \frac{5}{4} \cdot \frac{7}{6} \cdot \frac{11}{10} \cdot \frac{13}{12} = \frac{1001}{192} > 5$. By Problem 2(e), a product of sufficiently large powers of 2, 3, 5, 7, 11, and 13 will therefore have an abundancy arbitrarily close to a number greater than 5, thus there exists a product of powers of these six primes with abundancy greater than 5. Note that it is not necessary to find such a number because Problem 2(e) provides a *least* upper bound.

6. Let p_j be the jth prime, so $p_1 = 2$, $p_2 = 3$, and so on. Let P_k be the product $\displaystyle\prod_{j=1}^{k} \frac{p_j}{p_j - 1} = \prod_{j=1}^{k}\left(1 + \frac{1}{p_j - 1}\right)$. By Problem 2(e), a product of powers of the primes $p_1, p_2, \ldots, p_k$ will have abundancy less than or equal to P_k; in fact, because P_k is *least* upper bound, it follows that there exist integers whose abundance is arbitrarily close to P_k. Note that in general, if $a_1, a_2, \ldots, a_k$ are positive real numbers, then $\displaystyle\prod_{j=1}^{k}(1 + a_j) \geq \sum_{j=1}^{k} a_j$, because when the product on the left is expanded out, each a_j appears as a term in the product, and all other terms that appear are also positive. Thus $P_k = \displaystyle\prod_{j=1}^{k}\left(1 + \frac{1}{p_j - 1}\right) \geq \sum_{j=1}^{k} \frac{1}{p_j - 1} \geq \sum_{j=1}^{k} \frac{1}{p_j}$ for all k. By Fact 2, the sum on the right side grows without bound as k increases, and therefore P_k does as well. Because there exist integers with abundance arbitrarily close to P_k for each k, it follows that there are integers with arbitrarily large abundance, as desired.

7. Suppose that m and k are as given, and $A(n) = \frac{k}{m}$ for some n. Because $A(n) = \frac{\sigma(n)}{n}$ has denominator m when reduced, it follows that n must be a multiple of m. Alternatively for this conclusion, $\frac{\sigma(n)}{n} = \frac{k}{m}$ implies that $m\sigma(n) = kn$; it therefore follows that $m \mid kn$, and combined with the fact that m and k are relatively prime, this gives $m \mid n$. Next, note that $A(m) = \frac{\sigma(m)}{m}$, and by assumption, $\sigma(m) > k$, so $A(m) > \frac{k}{m}$. But m divides n, so by Problem 3, $A(m) \leq A(n)$. Thus $A(n) > \frac{k}{m}$, which contradicts the assumption that $A(n) = \frac{k}{m}$. Thus $\frac{k}{m}$ is an abundancy outlaw.

8. Examples include $\frac{5}{4}$, $\frac{7}{6}$, $\frac{11}{6}$, $\frac{9}{8}$, $\frac{11}{8}$, $\frac{13}{8}$, $\frac{10}{9}$, and $\frac{11}{9}$. To receive full credit, for each example $\frac{k}{m}$, it must be shown that $k < \sigma(m)$ and noted that k and m are relatively prime.

9. If n is an odd perfect number, then $A(n) = 2$ and 2 does not divide n. Thus by Problem 2(c), $A(2n) = A(2)A(n) = \frac{3}{2} \cdot 2 = 3$.

10. **a.** Assume that $A(n) = \frac{5}{3}$. Then n is a multiple of 3 because $3\sigma(n) = 5n$. If n is even, then $6 \mid n$, thus by Problem 3, $A(n) \geq A(6) = 2 > \frac{5}{3}$, a contradiction. Thus n is odd.

b. Because n is odd by Problem 10(a), and $3\sigma(n) = 5n$ is also odd, it follows that $\sigma(n)$ is odd. If 3 divides n (which was shown in the solution to 10(a)) but 9 does not, then by Fact 1, $\sigma(n)$ is a multiple of $\sigma(3) = 4$ and would therefore be even, a contradiction. Thus n is a multiple of 9.

c. Because $9 \mid n$ by Problem 10(b), if $5 \mid n$ then by Problem 3, $A(n) \geq A(9 \cdot 5)$, which by Problem 2(c) is $A(9)A(5) = \frac{13 \cdot 6}{9 \cdot 5} = \frac{26}{15} > \frac{5}{3}$, a contradiction. Thus n is not divisible by 5.

d. Recall that n is odd by Problem 10(a), thus $5n$ is odd. By Problem 10(c), n is not a multiple of 5 and hence relatively prime to 5, thus by Problem 2(c), $A(5n) = A(5)A(n) = \frac{6}{5} \cdot \frac{5}{3} = 2$, making $5n$ perfect. Thus $5n$ is an odd perfect number.

11. One possible answer is $n = 2^{11} \cdot 3^3 \cdot 5^2 \cdot 7^2 \cdot 13 \cdot 19 \cdot 31$. To find this, recall from Problem 2(a) that $A(p^k) = \dfrac{p^{k+1} - 1}{p^k(p-1)}$. Thus prime powers for which $p^{k+1} - 1$ is divisible by 5 (but $p - 1$ is not) are of particular interest; likely, more than one of them will need to appear in the prime factorization of n, because n will likely be divisible by 5, and $A(5^k)$ will have factors of 5 in its denominator. Prime powers of this form, which are candidates to appear in the prime factorization of n, are 2^3, 2^7, 2^{11}, 3^3, 7^3, 13^3, 17^3, and 19. Thus finding a possible value of n can be viewed as a "game" of adding additional prime power factors to n so that unwanted factors in the numerator or denominator of $A(n)$ are canceled out. Of the candidates considered so far:

- The number 2^3 is likely too small; the numerators of $A(p^k)$ for odd p are likely to contain factors of 2, so n will need to be divisible by a large power of 2 to cancel these out.

- For 2^7, note that $2^8 - 1 = 255 = 3 \cdot 5 \cdot 17$; it may or may not be convenient to have n divisible by 17.
- For 2^{11}, note that $2^{12} - 1 = 4095 = 3^2 \cdot 5 \cdot 7 \cdot 13$, which is promising because it has no large prime factors.
- For 3^3, note that $3^4 - 1 = 80 = 2^4 \cdot 5$, which seems promising.
- For 7^3, 13^3, and 17^3, the numerator $p^{k+1} - 1$ is divisible by several factors of 2, which may be hard to cancel out.

Thus it seems most promising, as a guess, to have n divisible by 2^7 or 2^{11}, 3^3, and 19. This puts three factors of 5 in the numerator of $A(n)$, so n must also be divisible by 5^2; further, because $A(5^2)$ has 31 in the numerator, n must be divisible by 31. Then note that

- $A(2^7 \cdot 3^3 \cdot 5^2 \cdot 19 \cdot 31) = \frac{255}{128} \cdot \frac{40}{27} \cdot \frac{31}{25} \cdot \frac{20}{19} \cdot \frac{32}{31} = \frac{2^3 \cdot 5 \cdot 17}{3^2 \cdot 19}$;
- $A(2^{11} \cdot 3^3 \cdot 5^2 \cdot 19 \cdot 31) = \frac{4095}{2048} \cdot \frac{40}{27} \cdot \frac{31}{25} \cdot \frac{20}{19} \cdot \frac{32}{31} = \frac{5 \cdot 7 \cdot 13}{2 \cdot 3 \cdot 19}$.

In the first case, there is no way to remove the remaining 2s from the numerator, but the second possibility remains viable.

In the second case, n must be divisible by 7 (to remove the 7 from the numerator), but $A(7) = \frac{8}{7}$, which would add too many factors of 2 to the numerator. However, $A(7^2) = \frac{3 \cdot 19}{7^2}$, which will take care of the 19 in the denominator of $A(n)$. That is,

$$A(2^{11} \cdot 3^3 \cdot 5^2 \cdot 7^2 \cdot 19 \cdot 31) = \frac{5 \cdot 7 \cdot 13}{2 \cdot 3 \cdot 19} \cdot \frac{3 \cdot 19}{7^2} = \frac{5 \cdot 13}{2 \cdot 7}.$$

Finally, $A(13) = \frac{2 \cdot 7}{13}$, from which

$$A(2^{11} \cdot 3^3 \cdot 5^2 \cdot 7^2 \cdot 13 \cdot 19 \cdot 31) = \frac{5 \cdot 13}{2 \cdot 7} \cdot \frac{2 \cdot 7}{13} = 5,$$

as desired.

6 Individual Round

Problem 1. Given that x and y are real numbers such that $x^2 + y^2 = 202$ and $x - y = 6$, compute $|x + y|$.

Problem 2. Let P be a polynomial each of whose coefficients is either 0, 2, or 6. Given that $P(4) = 2026$, compute $P(10)$.

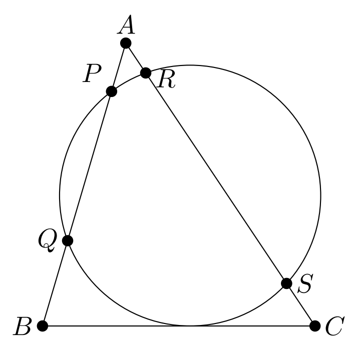
Problem 3. Jannik rolls two standard fair six-sided dice and tells Carlos the product of the two numbers rolled. Carlos guesses the sum of the two numbers rolled, by choosing the most likely sum based on the product Jannik told him. Compute the probability that Carlos guesses correctly.

Problem 4. Compute $\sin^2 42° + \sin^2 78° - \sin 42° \sin 78°$.

Problem 5. Compute the greatest possible number of sides of a convex polygon whose angle measures (in degrees) in some order are distinct consecutive integers.

Problem 6. Suppose that $\triangle PQR$ has $PQ = QR = 2026$ and $PR = 4051$. Let $\triangle ABC$ have integer side lengths and $\text{m}\angle C \leq \text{m}\angle R$. Compute the minimum possible perimeter of $\triangle ABC$.

Problem 7. In triangle ABC, a circle that is tangent to side $\overline{BC}$ intersects side $\overline{AB}$ at points P and Q, where P is the point closer to A, and intersects side $\overline{AC}$ at points R and S, where R is the point closer to A, as shown. Given that $AB = 13$, $AC = 15$, $AP = 4$, $AQ = 9$, and $AR = 3$, compute BC.



Problem 8. The 60 students of the West Coast ARML team and the 120 students of the East Coast ARML team are at a party. A total of 1800 handshakes occur between a West Coast student and an East Coast student, with no pair of people shaking hands more than once and no other handshakes occurring. For each pair of students at the party, the coaches record how many students shook hands with both. Compute the minimum possible average of the $\binom{180}{2}$ numbers recorded by the coaches.

Problem 9. Compute the second least positive integer n such that $n + 1$ and $5n + 1$ are perfect squares.

Problem 10. Let $\triangle ABC$ be a triangle with incenter I and incircle ω which is tangent to sides $\overline{BC}$, $\overline{CA}$, and $\overline{AB}$ at D, E, and F, respectively. Let $\overline{TD}$ be a diameter of ω. Suppose that the angle bisector of $\angle BAC$ splits $\triangle ABC$ into two triangles, one of which is isosceles with vertex angle at A. Given that $TD = 4$ and $[FIT] - [TIE] = \frac{35}{27}$, compute $AI \cdot [FIE]$.

7 Individual Round Answers

Answer 1. $4\sqrt{23}$

Answer 2. 66622

Answer 3. $\dfrac{31}{36}$

Answer 4. $\dfrac{3}{4}$ (or 0.75)

Answer 5. 16

Answer 6. 93

Answer 7. 12

Answer 8. $\dfrac{430}{179}\left(\text{or } 2\dfrac{72}{179}\right)$

Answer 9. 24

Answer 10. $\dfrac{70-2\sqrt{141}}{9}$

8 Individual Round Solutions

Problem 1. Given that x and y are real numbers such that $x^2 + y^2 = 202$ and $x - y = 6$, compute $|x + y|$.

Solution 1. Note that $(x - y)^2 = x^2 - 2xy + y^2 = 36$. Because $x^2 + y^2 = 202$, it follows that $2xy = 166$. Then note that $(x + y)^2 = x^2 + 2xy + y^2 = 202 + 166 = 368$, hence $|x + y| = \sqrt{368} = \mathbf{4\sqrt{23}}$.

Problem 2. Let P be a polynomial each of whose coefficients is either 0, 2, or 6. Given that $P(4) = 2026$, compute $P(10)$.

Solution 2. Let $Q(x) = \frac{1}{2}P(x)$, so $Q(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$, where $a_i \in \{0, 1, 3\}$ for all $0 \le i \le n$. Then $Q(4) = a_n 4^n + a_{n-1}4^{n-1} + \cdots + a_0 = 1013$. Note that $a_0 \equiv 1013 \bmod 4$, and hence $a_0 = 1$. Similarly, $4a_1 + a_0$ and $16a_2 + 4a_1 + a_0$ are congruent to 1013 modulo 16 and 64, respectively, which yields $a_1 = 1$ and $a_2 = 3$. Continuing in this fashion, note that this process is precisely computing the base-4 representation of $P(4)$; that is, the coefficients of Q will be the digits of 1013 when written in base 4. Because $1013 = 33311_4$, it follows that

$$Q(x) = 3x^4 + 3x^3 + 3x^2 + x + 1,$$

hence $P(10) = 2 \cdot Q(10) = \mathbf{66622}$.

Problem 3. Jannik rolls two standard fair six-sided dice and tells Carlos the product of the two numbers rolled. Carlos guesses the sum of the two numbers rolled, by choosing the most likely sum based on the product Jannik told him. Compute the probability that Carlos guesses correctly.

Solution 3. If the product is a number that can be expressed in only one way as the product of two factors no greater than 6, then Carlos can determine what the two die numbers are and will be able to guess the sum precisely. If the product is 4 ($2 \cdot 2$ or $1 \cdot 4$) or 6 ($2 \cdot 3$ or $1 \cdot 6$) or 12 ($2 \cdot 6$ or $3 \cdot 4$), then Carlos should guess one of the possible sums. Either sum is equally likely, except in the case where the product is 4. Then Carlos should guess that the sum is 5, not 4, because it is twice as likely that the rolls are 1 and 4, rather than 2 and 2. Those three cases occur with probabilities $\frac{3}{36}$, $\frac{4}{36}$, and $\frac{4}{36}$, respectively, with Carlos guessing correctly $\frac{2}{3}$, $\frac{1}{2}$, and $\frac{1}{2}$ of the time. The remaining $\frac{25}{36}$ of the time, Carlos always guesses correctly. Thus Carlos can guess correctly with probability $\mathbf{\frac{31}{36}}$.

Problem 4. Compute $\sin^2 42^\circ + \sin^2 78^\circ - \sin 42^\circ \sin 78^\circ$.

Solution 4. Note that the given expression equals $\sin^2(60^\circ - A) + \sin^2(60^\circ + A) - \sin(60^\circ - A)\sin(60^\circ + A)$, where $A = 18^\circ$. By a double-angle identity and then addition/subtraction identities,

$$\begin{aligned}
\sin^2(60^\circ - A) + \sin^2(60^\circ + A) &= \frac{1}{2}\big((1 - \cos(120^\circ - 2A)) + (1 - \cos(120^\circ + 2A))\big) \\
&= 1 - \frac{\cos(120^\circ - 2A) + \cos(120^\circ + 2A)}{2} \\
&= 1 - \frac{(\cos 120^\circ \cos 2A + \sin 120^\circ \sin 2A) + (\cos 120^\circ \cos 2A - \sin 120^\circ \sin 2A)}{2} \\
&= 1 + \frac{\cos 2A}{2}.
\end{aligned}$$

By a product-to-sum identity,

$$\begin{aligned}\sin(60^\circ - A)\sin(60^\circ + A) &= \frac{\cos(-2A) - \cos 120^\circ}{2} \\ &= \frac{\cos 2A}{2} + \frac{1}{4}.\end{aligned}$$

Hence the required expression is $1 + \frac{\cos 2A}{2} - (\frac{\cos 2A}{2} + \frac{1}{4}) = \mathbf{\frac{3}{4}}$ (independent of A).

Alternate Solution: Let $x = \sin 42^\circ$, $y = \sin 78^\circ$, $a = \cos 42^\circ$, and $b = \cos 78^\circ$. Then note that $\sin 120^\circ = \sin(42^\circ + 78^\circ) = bx + ay = \frac{\sqrt{3}}{2}$, and $\cos 120^\circ = ab - xy = -\frac{1}{2}$. Square both equations to obtain

$$\begin{aligned}b^2x^2 + 2abxy + a^2y^2 &= \frac{3}{4} \\ a^2b^2 - 2abxy + x^2y^2 &= \frac{1}{4}.\end{aligned}$$

Subtract these two equations and factor to obtain

$$(x^2 - a^2)(b^2 - y^2) + 4abxy = \frac{1}{2}.$$

Because $a^2 + x^2 = b^2 + y^2 = 1$, it follows that

$$(2x^2 - 1)(1 - 2y^2) + 4xy\left(xy - \frac{1}{2}\right) = \frac{1}{2},$$

or equivalently, $2x^2 + 2y^2 - 1 - 2xy = \frac{1}{2}$, from which $x^2 + y^2 - xy = \frac{3}{4}$. Note that this would still hold if 42° and 78° were replaced by any two angles summing to 120°.

Alternate Solution: Let x and y be the values of the angles 42° and 78° in radians, respectively. The desired quantity is $\dfrac{\sin^3 x + \sin^3 y}{\sin x + \sin y}$. Let $\alpha = e^{ix}$ and $\beta = e^{iy}$. Then

$$\sin x = \frac{1}{2i}(e^{ix} - e^{-ix}) = \frac{1}{2i}\left(\alpha - \frac{1}{\alpha}\right) \quad \text{and} \quad \sin y = \frac{1}{2i}\left(\beta - \frac{1}{\beta}\right).$$

Then note that $x + y = \frac{2\pi}{3}$, so $\alpha\beta = e^{i\cdot\frac{2\pi}{3}}$, which is a cube root of unity, thus $(\alpha\beta)^3 = 1$ and $\alpha^3 = \frac{1}{\beta^3}$. This implies

$$\begin{aligned}\sin^3 x + \sin^3 y &= -\frac{1}{8i}\left[\left(\alpha - \frac{1}{\alpha}\right)^3 + \left(\beta - \frac{1}{\beta}\right)^3\right] \\ &= -\frac{1}{8i}\left(\alpha^3 - 3\alpha + \frac{3}{\alpha} - \frac{1}{\alpha^3} + \beta^3 - 3\beta + \frac{3}{\beta} - \frac{1}{\beta^3}\right) \\ &= -\frac{1}{8i}\left(-3\alpha + \frac{3}{\alpha} - 3\beta + \frac{3}{\beta}\right) \\ &= \frac{3}{8i}\left(\left(\alpha - \frac{1}{\alpha}\right) + \left(\beta + \frac{1}{\beta}\right)\right) \\ &= \frac{3}{4}(\sin x + \sin y).\end{aligned}$$

Thus the desired quantity is $\frac{3}{4}$, and in fact is independent of the actual angles, provided that their sum is 120°.

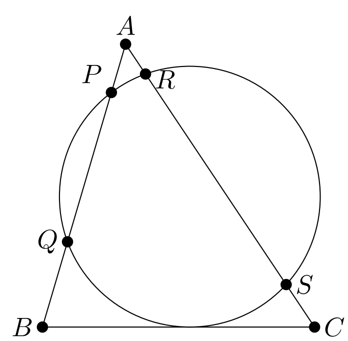
Problem 5. Compute the greatest possible number of sides of a convex polygon whose angle measures (in degrees) in some order are distinct consecutive integers.

Solution 5. Let n be the number of sides of the polygon. In degrees, the average angle will be $a = 180 - \frac{360}{n}$, so that will be the median of the list of consecutive integer measures. If a is an integer, then n must be odd, and the angles will be a, $a \pm 1$, $a \pm 2$, $\ldots$, $a \pm \frac{n-1}{2}$. For convexity, $180 > a + \frac{n-1}{2} = 180 - \frac{360}{n} + \frac{n-1}{2}$, which simplifies to $n(n-1) < 720$, so $n \leq 27$ is a divisor of 360. The odd divisors of 360 are divisors of 45, and the greatest of these less than 27 is 15. However, there is also the possibility that n is even and a is a half-integer, in which case the angles will be $a \pm \frac{1}{2}, a \pm \frac{3}{2}, \ldots, a \pm \frac{n-1}{2}$. For convexity, $180 > 180 - \frac{360}{n} + \frac{n-1}{2}$, which again simplifies to $n(n-1) < 720$, so $n \leq 27$ and $\frac{360}{n}$ is a half-integer, thus n is 16 times an odd divisor of 45. The only such number is **16**, which is greater than 15 and is thus the greatest possible number of sides.

Problem 6. Suppose that $\triangle PQR$ has $PQ = QR = 2026$ and $PR = 4051$. Let $\triangle ABC$ have integer side lengths and $\mathrm{m}\angle C \leq \mathrm{m}\angle R$. Compute the minimum possible perimeter of $\triangle ABC$.

Solution 6. Let $a = BC$, $b = AC$, and $c = AB$. Assume, without loss of generality, that $\angle C$ is the smallest angle in $\triangle ABC$, and then that $\angle A$ is the largest, so that $c \leq b \leq a$. By the Law of Cosines, $\cos C = \frac{a^2+b^2-c^2}{2ab}$, and reducing c while holding a and b fixed will always (provided this does not violate the Triangle Inequality) result in a triangle with smaller perimeter and a smaller $\angle C$, so assume that $b + c = a + 1$. The perimeter of ABC is $a + b + c = 2a + 1$, so it suffices to minimize a. Substitute for c in the Law of Cosines, obtaining $\cos C = \frac{a^2+b^2-(a-b+1)^2}{2ab}$, which simplifies to $1 + \frac{1}{a} - \frac{1}{b} - \frac{1}{2ab}$. It follows that $\cos C \leq 1 - \frac{1}{2a^2}$ because $b \leq a$. Applying the Law of Cosines to $\triangle PQR$ gives $\cos R = \frac{2026^2+4051^2-2026^2}{2\cdot 2026\cdot 4051} = \frac{4051}{4052}$. If $\angle C \leq \angle R$, then $\cos C \geq \cos R$, so $\frac{1}{2a^2} \leq \frac{1}{4052}$. This gives $a^2 \geq 2026$, and because a is an integer, $a \geq 46$. The minimum possible perimeter of $\triangle ABC$ is then $2 \cdot 46 + 1 = \mathbf{93}$.

Problem 7. In triangle ABC, a circle that is tangent to side $\overline{BC}$ intersects side $\overline{AB}$ at points P and Q, where P is the point closer to A, and intersects side $\overline{AC}$ at points R and S, where R is the point closer to A, as shown. Given that $AB = 13$, $AC = 15$, $AP = 4$, $AQ = 9$, and $AR = 3$, compute BC.



Solution 7. By the Power of a Point Theorem, $AR \cdot AS = AP \cdot AQ$, so $3AS = 4 \cdot 9$ and $AS = 12$. Hence $BQ = 13 - 9 = 4$ and $CS = 15 - 12 = 3$. Let D be the point where the circle intersects BC. Applying Power of a Point from B and C, respectively, $BD^2 = BQ \cdot BP$, so $BD = \sqrt{4 \cdot 9} = 6$ and $CD^2 = CS \cdot CR$, so $CD = \sqrt{3 \cdot 12} = 6$. Thus $BC = BD + CD = 6 + 6 = \mathbf{12}$.

Problem 8. The 60 students of the West Coast ARML team and the 120 students of the East Coast ARML team are at a party. A total of 1800 handshakes occur between a West Coast student and an East Coast student, with no pair of people shaking hands more than once and no other handshakes occurring. For each pair of students at the party, the coaches record how many students shook hands with both. Compute the minimum possible average of the $\binom{180}{2}$ numbers recorded by the coaches.

Solution 8. It suffices to compute the sum of the numbers recorded; that is, the sum over all pairs of students of the number of students that shook hands with both. Consider the number of times each student is counted in this sum. If student s shook hands with d_s people, then s is counted once in $\binom{d_s}{2}$ different terms of the sum. Thus the sum is equal to

$$\sum_s \binom{d_s}{2},$$

where s ranges across all students at the party. Break the sum up into two parts, one sum over the West Coast (WC) students and one sum over the East Coast (EC) students. The sum over the West Coast is

$$\sum_{s\in\mathrm{WC}} \binom{d_s}{2} = \frac{1}{2}\sum_{s\in\mathrm{WC}} d_s^2 - \frac{1}{2}\sum_{s\in\mathrm{WC}} d_s.$$

By the Cauchy–Schwarz Inequality, it follows that

$$\left(\sum_{s\in\mathrm{WC}} 1\right)\left(\sum_{s\in\mathrm{WC}} d_s^2\right) \geq \left(\sum_{s\in\mathrm{WC}} d_s\right)^2,$$

so

$$\sum_{s\in\mathrm{WC}} \binom{d_s}{2} \geq \frac{1}{2\cdot 60}\left(\sum_{s\in\mathrm{WC}} d_s\right)^2 - \frac{1}{2}\sum_{s\in\mathrm{WC}} d_s.$$

By grouping the handshakes by which West Coast student is involved, observe that

$$\sum_{s\in\mathrm{WC}} d_s$$

is precisely the total number of handshakes, which is 1800. It follows that

$$\sum_{s\in\mathrm{WC}} \binom{d_s}{2} \geq \frac{1800^2}{2\cdot 60} - \frac{1800}{2} = 900\cdot 29.$$

Similarly, it follows that

$$\sum_{s\in\mathrm{EC}} \binom{d_s}{2} \geq \frac{1800^2}{2\cdot 120} - \frac{1800}{2} = 900\cdot 14.$$

Thus the average number recorded is

$$\frac{1}{\binom{180}{2}}\sum_s \binom{d_s}{2} \geq \frac{1}{90\cdot 179}(900\cdot 29 + 900\cdot 14) = \frac{430}{179}.$$

It remains to check that this is attainable. Equality in the Cauchy–Schwarz applications requires d_s to be constant across all West Coast students and constant across all East Coast students. Divide the party into 4 groups, each with 15 West Coast students and 30 East Coast students. Within each group, suppose each of the $15\cdot 30$ possible handshakes occur. Then 1800 total handshakes occur. Then d_s is 30 for each of the West Coast students and 15 for each of the East Coast students, which gives an equality case. Thus the minimum possible average is $\boldsymbol{\frac{430}{179}}$.

Problem 9. Compute the second least positive integer n such that $n+1$ and $5n+1$ are perfect squares.

Solution 9. Do casework on the value of $n+1$. The least positive integer n such that $n+1$ is a perfect square is 3. For this value, $5\cdot 3+1 = 4^2$. The number $n+1$ is also a perfect square for $n = 8, 15, 24, 35, \ldots$, and the respective values of $5n+1$ are $41, 76, 121, 176, \ldots$. Noting that $121 = 11^2$, it follows that $n = \mathbf{24}$ is the desired value of n.

Alternate Solution: Let $n+1 = x^2$ and $5n+1 = y^2$ for positive integers x and y. Noting that $n = x^2 - 1$ from the first equation, this can be substituted into the second equation to get $5(x^2-1)+1 = y^2$ or $y^2 - 5x^2 = -4$. The *Lucas numbers* L_n, defined by $L_0 = 2$, $L_1 = 1$, and $L_n = L_{n-1} + L_{n-2}$ for $n \geq 2$ and the *Fibonacci numbers* F_n, defined by $F_0 = 1$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$ have the relationship $L_n^2 - 5F_n^2 = 4(-1)^n$, meaning that the solutions to this problem correspond to the odd-indexed values of these sequences. The first pairs are $(1,1)$, $(2,4)$, and $(5,11)$, and the corresponding values of n are 0, 3, and **24**.

Problem 10. Let $\triangle ABC$ be a triangle with incenter I and incircle ω which is tangent to sides $\overline{BC}$, $\overline{CA}$, and $\overline{AB}$ at D, E, and F, respectively. Let $\overline{TD}$ be a diameter of ω. Suppose that the angle bisector of $\angle BAC$ splits $\triangle ABC$ into two triangles, one of which is isosceles with vertex angle at A. Given that $TD = 4$ and $[FIT] - [TIE] = \frac{35}{27}$, compute $AI \cdot [FIE]$.

Solution 10. Because $BDIF$ is a cyclic quadrilateral (two opposite angles of $90°$), it follows that $\angle DIF = 180° - B$, so $\angle TIF = B$. Similarly, $\angle TIE = C$. Let r be the radius of ω. Then $r = \frac{TD}{2} = 2$, and

$$[FIT] - [TIE] = \frac{1}{2}r^2 \sin B - \frac{1}{2}r^2 \sin C = r^2 \cos \frac{B+C}{2} \sin \frac{B-C}{2} = r^2 \sin \frac{A}{2} \sin \frac{B-C}{2}.$$

Thus $\sin \frac{A}{2} \sin \frac{B-C}{2} = \frac{35}{108}$. Let the angle bisector of $\angle BAC$ meet $\overline{BC}$ at P. If $\triangle ABP$ is isosceles, then $\angle APB = B$. But $\angle APB = \angle PAC + \angle ACP = \frac{A}{2} + C$, so it follows that $B - C = \frac{A}{2}$. Similarly, if $\triangle ACP$ is isosceles, then $C - B = \frac{A}{2}$. From $\sin \frac{A}{2} \sin \frac{B-C}{2} = \frac{35}{108}$, it must be the case that $\sin \frac{B-C}{2}$ is positive, so $B - C$ is also positive. Thus $B - C = \frac{A}{2}$, so $\sin \frac{A}{2} \sin \frac{A}{4} = \frac{35}{108}$. Then

$$\frac{35}{108} = \sin \frac{A}{2} \sin \frac{A}{4} = 2\sin^2 \frac{A}{4} \cos \frac{A}{4} = 2\left(1 - \cos^2 \frac{A}{4}\right) \cos \frac{A}{4},$$

so $\cos \frac{A}{4} - \cos^3 \frac{A}{4} = \frac{35}{216}$. One solution to $x - x^3 = \frac{35}{216}$ is $x = \frac{1}{6}$. This implies that the other two solutions satisfy $x^2 + \frac{1}{6}x - \frac{35}{36} = 0$, so they are $x = \frac{-1 \pm \sqrt{141}}{12}$. Observe that $0° < \frac{A}{4} < 45°$, so $\frac{\sqrt{2}}{2} < \cos \frac{A}{4} < 1$. The values $\frac{1}{6}$ and $\frac{-1-\sqrt{141}}{12}$ are too small to be $\cos \frac{A}{4}$, so it must be that $\cos \frac{A}{4} = \frac{-1+\sqrt{141}}{12}$. Now compute

$$\begin{aligned} AI \cdot [FIE] &= \frac{r}{\sin \frac{A}{2}} \cdot \frac{1}{2}r^2 \sin(180° - A) = \frac{r^3 \sin A}{2 \sin \frac{A}{2}} \\ &= r^3 \cos \frac{A}{2} = r^3 \left(2\cos^2 \frac{A}{4} - 1\right) = 8\left(\frac{35 - \sqrt{141}}{36}\right) \\ &= \boldsymbol{\frac{70 - 2\sqrt{141}}{9}}. \end{aligned}$$

9 Relay Round

Problem 1-1. Compute the least positive integer n such that 6400 divides $n!$.

Problem 1-2. *Let T be the number you will receive.* Suppose that $\tan\theta = \frac{1}{T}$. Compute $\sin(2\theta)$.

Problem 1-3. *Let T be the number you will receive.* Suppose that $\sin\theta = \sqrt{|T|}$ for some $0^\circ < \theta < 90^\circ$. Compute $1 + (1-\cos\theta) + (1-\cos\theta)^2 + (1-\cos\theta)^3 + \cdots$.

Problem 2-1. Compute the number of ordered pairs of positive integers (m, n) such that $\sqrt{m^2 + \sqrt{n}} = 20\sqrt{26}$.

Problem 2-2. *Let T be the number you will receive.* Compute the positive difference of the two real solutions to the equation $|3x+2| + |2x+3| = \left\lfloor \frac{T}{2} \right\rfloor$.

Problem 2-3. *Let T be the number you will receive.* Compute the number of infinite geometric progressions of nonzero integers for which there exists a term that is T greater than the previous term.

10 Relay Round Answers

Answer 1-1. 10

Answer 1-2. $\dfrac{20}{101}$

Answer 1-3. $\dfrac{\sqrt{101}}{9}$

Answer 2-1. 101

Answer 2-2. 20

Answer 2-3. 16

11 Relay Round Solutions

Problem 1-1. Compute the least positive integer n such that 6400 divides $n!$.

Solution 1-1. The prime factorization of 6400 is $2^8 \cdot 5^2$, meaning n needs to be at least 10 for $n!$ to have sufficient multiples of 5 (one at 5, one at 10), and at least 10 for $n!$ to have sufficient multiples of 2 (one at 2, two at 4, one at 6, three at 8, one at 10). The answer is **10** (of note, $10!/6400 = 567$).

Alternate Solution: Note that $6400 = 2^8 \cdot 5^2$, so $n \geq 10$ because 2 factors of 5 are needed. With $n = 10$, the greatest power of 2 that divides $n!$ is $2^1 \cdot 2^2 \cdot 2^1 \cdot 2^3 \cdot 2^1 = 2^8$, so $6400 \mid 10!$ and $6400 \nmid 9!$, and the answer is **10**.

Problem 1-2. *Let T be the number you will receive.* Suppose that $\tan\theta = \frac{1}{T}$. Compute $\sin(2\theta)$.

Solution 1-2. Note that $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\pm\sqrt{1-\cos^2\theta}}{\cos\theta} = \frac{1}{T}$, thus $\cos^2\theta = \frac{T^2}{T^2+1}$. On the other hand,

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{(\sin\theta)(\cos\theta)}{\cos^2\theta} = \frac{\sin 2\theta}{2\cos^2\theta},$$

hence $\sin 2\theta = (\tan\theta)(2\cos^2\theta) = \frac{2T}{T^2+1}$. With $T = 10$, the answer is $\mathbf{\frac{20}{101}}$.

Problem 1-3. *Let T be the number you will receive.* Suppose that $\sin\theta = \sqrt{|T|}$ for some $0^\circ < \theta < 90^\circ$. Compute $1 + (1-\cos\theta) + (1-\cos\theta)^2 + (1-\cos\theta)^3 + \cdots$.

Solution 1-3. Note that the given sum is an infinite series with first term 1 and common ratio $1-\cos\theta$, and therefore is equal to $\frac{1}{1-(1-\cos\theta)} = \frac{1}{\cos\theta}$. However, $\cos\theta = \sqrt{1-|T|}$ so the given sum is $\frac{1}{\sqrt{1-|T|}}$. Because $T = \frac{20}{101}$, it follows that the given sum equals $\mathbf{\frac{\sqrt{101}}{9}}$.

Problem 2-1. Compute the number of ordered pairs of positive integers (m, n) such that $\sqrt{m^2 + \sqrt{n}} = 20\sqrt{26}$.

Solution 2-1. Square both sides of the given equation to obtain $m^2 + \sqrt{n} = 400 \cdot 26 = 10400$. Because m is an integer, it follows that $\sqrt{n}$ must also be an integer, and so count the number of perfect squares between 1 and 10400. Because $101^2 = 10201$ and $102^2 = 10404$, and because each perfect square m^2 determines a unique n, the answer is **101**.

Problem 2-2. *Let T be the number you will receive.* Compute the positive difference of the two real solutions to the equation $|3x+2| + |2x+3| = \left\lfloor \frac{T}{2} \right\rfloor$.

Solution 2-2. Consider three cases:

(i) If $x \geq -\frac{2}{3}$ then $3x+2$ and $2x+3$ are both nonnegative, and the given equation can be written as $5x + 5 = \left\lfloor \frac{T}{2} \right\rfloor$, whose solution is $x = \frac{\lfloor T/2 \rfloor}{5} - 1$.

(ii) If $-\frac{3}{2} < x < -\frac{2}{3}$, then $3x+2 < 0$, while $2x+3 > 0$, and the given equation can be written as $-x+1 = \left\lfloor \frac{T}{2} \right\rfloor$, whose solution is $x = 1 - \left\lfloor \frac{T}{2} \right\rfloor$.

(iii) If $x \leq -\frac{3}{2}$, then $3x+2$ and $2x+3$ are both nonpositive, and the given equation can be written as $-5x - 5 = \left\lfloor \frac{T}{2} \right\rfloor$, whose solution is $x = \frac{\lfloor T/2 \rfloor}{-5} - 1$.

With $T = 101$, in case (i), $x = 9$ (and $x \geq -\frac{2}{3}$, as required). In case (ii), $x = -49$, but -49 does not lie in the interval $(-\frac{3}{2}, -\frac{2}{3})$, so there are no solutions for this case. Finally, in case (iii), $x = -11$ (and $x \leq -\frac{3}{2}$, as required). The desired difference is $|9 - (-11)| = \mathbf{20}$.

Problem 2-3. *Let T be the number you will receive.* Compute the number of infinite geometric progressions of nonzero integers for which there exists a term that is T greater than the previous term.

Solution 2-3. Let the geometric progression have first term a and ratio r. Observe that a and r are both nonzero integers, because if r was not an integer, then the progression would eventually contain nonintegers. Let the term in question be ar^n, where $n \geq 1$ so that $ar^n - ar^{n-1} = T$. Then $ar^{n-1}(r-1) = T$, so $r-1$ divides T. With $T = 20$, there are 12 possible values of $r-1$ (and hence r). The table below lists the possible values, alongside the possible values of n (which are determined by the condition that r^{n-1} divides $\frac{20}{r-1}$).

$r-1$	r	possible n
1	2	1, 2, 3
2	3	1
4	5	1, 2
5	6	1
10	11	1
20	21	1
-1	0	none
-2	-1	all
-4	-3	1
-5	-4	1, 2
-10	-9	1
-20	-19	1

Except for $r = -1$, there are 14 possible geometric progressions. With $r = -1$, there are 2 progressions: $10, -10, 10, -10, \ldots$ and $-10, 10, -10, 10, \ldots$. Thus there are a total of **16** progressions that satisfy the given conditions.

12 Super Relay

1. As x ranges over the real numbers, compute the least possible value of $|x-20|+|x-26|$.

2. Let $T = \text{TNYWR}$. Compute the units digit of 2026^{T-6}.

3. Let $T = \text{TNYWR}$. Compute the value of y that satisfies the system of simultaneous equations

$$\begin{aligned} x^2 - y^2 &= -3T^2 \\ x + y &= T. \end{aligned}$$

4. Let $T = \text{TNYWR}$. A box has 50 marbles; 25 of them are red, and the other 25 are blue. Frank makes exactly T *drawings*, where a drawing consists of the following. He chooses a marble from the box at random, records its color, then **replaces** the marble back into the box. Compute the probability that Frank's list of T colors contains *both* the colors red and blue.

5. Let $T = \text{TNYWR}$. Given that $\sin\theta = T$, compute the sum of the infinite series $\sin\theta + \sin^2\theta + \sin^3\theta + \sin^4\theta + \cdots$.

6. Let $T = \text{TNYWR}$. A circle of area $T\pi$ is inscribed in a square of side length s. Compute s.

7. Let $T = \text{TNYWR}$, and let $b = T^3$. Compute the value of x for which $\log_b(x+T) = \log_T(T+1)$.

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15. A (nondegenerate) triangle has sides of length 20, 26, and x, where x is a positive integer. Compute the number of possible values of x.

14. Let $T = \text{TNYWR}$. Compute $\operatorname{lcm}(20, 26, T)$.

13. Let $T = \text{TNYWR}$. Isosceles trapezoid $ABCD$ has a perimeter of $\frac{T}{10}$, $AB = 20$, $CD = 26$, and $\overline{AB} \parallel \overline{CD}$. Let M and N be the respective midpoints of $\overline{AD}$ and $\overline{BC}$. Compute $AM + MN - NC$.

12. Let $T = \text{TNYWR}$. Let $i = \sqrt{-1}$, and for positive integers n, let $S_n = 1 + 2 + 3 + \cdots + (n-1) + n$. Compute the number of integers k such that $1 \le k \le 50$ and $i^T = i^{S_k}$.

11. Let $T = \text{TNYWR}$. Compute the sum of all real number(s) x that satisfy $\sqrt{x+T+4} = x - T$.

10. Let $T = \text{TNYWR}$, and let $J = 10T - 10$. In triangle LEO, $\angle E$ is a right angle, $EO = J$, and $LO = J + 1$. Compute LE.

9. Let $T = \text{TNYWR}$. Speed reader Vince reads the dictionary at an impressive rate of T pages per 5 minutes. At this rate, after h hours have elapsed, where h is an integer, Vince will have <u>first</u> read at least 2026 pages of the dictionary. Compute h.

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8. Let A be the number you will receive from position 7, and let B be the number you will receive from position 9. Let $M = \max\left\{\sqrt{A}, \sqrt{B}\right\}$, and let $m = \min\left\{\sqrt{A}, \sqrt{B}\right\}$. Two spheres are tangent and have radii M and m. Let d_{max} be the greatest possible distance between the centers of these spheres, and let d_{min} be the least possible distance between the centers of these spheres. Compute $d_{\text{max}} \cdot d_{\text{min}}$.

13 Super Relay Answers

1. 6

2. 1

3. 2

4. $\dfrac{1}{2}$

5. 1

6. 2

7. 25

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15. 39

14. 780

13. 23

12. 13

11. 19

10. 19

9. 9

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8. 16

14 Super Relay Solutions

Problem 1. As x ranges over the real numbers, compute the least possible value of $|x-20|+|x-26|$.

Solution 1. More generally, let $f(x)=|x-a|+|x-b|$, where a and b are positive reals and $b>a$. If $x>b$, then $f(x)=(x-a)+(x-b)=2x-(a+b)>b-a$. If $a\leq x\leq b$, then $f(x)=(b-x)+(a-x)=b-a$. Finally, if $x<a$, then $f(x)=(a-x)+(b-x)=a+b-2x>b-a$. Thus the desired minimum value is $b-a$. With $b=26$ and $a=20$, the answer is **6**.

Problem 2. Let $T=$ TNYWR. Compute the units digit of 2026^{T-6}.

Solution 2. Let $K=T-6$. If K is a positive integer, then 2026^K always has a units digit of 6. However, if $K=0$, then the units digit of 2026^0 is 1. Because $T=6$, $K=0$, hence the answer is **1**.

Problem 3. Let $T=$ TNYWR. Compute the value of y that satisfies the system of simultaneous equations

$$x^2-y^2=-3T^2$$
$$x+y=T.$$

Solution 3. Rewrite the first given equation as $(x+y)(x-y)=-3T^2$. Assuming $T\neq 0$, substitute T for $x+y$ to give $x-y=-3T$. Thus $y=\frac{1}{2}\big((x+y)-(x-y)\big)=2T$. With $T=1$, the desired value of y is **2**.

Problem 4. Let $T=$ TNYWR. A box has 50 marbles; 25 of them are red, and the other 25 are blue. Frank makes exactly T *drawings*, where a drawing consists of the following. He chooses a marble from the box at random, records its color, then **replaces** the marble back into the box. Compute the probability that Frank's list of T colors contains *both* the colors red and blue.

Solution 4. Because there are equally as many red as blue marbles, in any drawing, the probability of choosing a red marble equals the probability of choosing a blue marble, and this common probability is $\frac{1}{2}$. Suppose that Frank chooses color $\mathcal{C}$ on the first drawing. Because replacement occurs and the probabilities of drawing a particular color are independent, the probability that **all** T drawings are $\mathcal{C}$ is $(\frac{1}{2})^{T-1}$. If this event does *not* occur, then the T drawings will include *both* colors, and this event will occur with probability $1-(\frac{1}{2})^{T-1}$. With $T=2$, the desired probability is therefore $\boldsymbol{\frac{1}{2}}$.

Problem 5. Let $T=$ TNYWR. Given that $\sin\theta=T$, compute the sum of the infinite series $\sin\theta+\sin^2\theta+\sin^3\theta+\sin^4\theta+\cdots$.

Solution 5. The infinite geometric series $S=a+ar^1+ar^2+\cdots$ has sum $\frac{a}{1-r}$, provided that $|r|<1$. Given that $T=\sin\theta$, the desired series is geometric with $a=T$ and $r=T$. As long as $|T|<1$, the desired sum is $\frac{T}{1-T}$. With $T=\frac{1}{2}$, this sum is $\frac{1/2}{1-1/2}=\mathbf{1}$. (It turns out that the exact value of θ can be determined as $\frac{\pi}{2}\pm\frac{\pi}{3}+2\pi k$, where k is any integer, but in this case, this is irrelevant to computing the sum of the given series.)

Problem 6. Let $T=$ TNYWR. A circle of area $T\pi$ is inscribed in a square of side length s. Compute s.

Solution 6. The circle's radius is $\sqrt{T}$. Therefore $s=2\sqrt{T}$. With $T=1$, the answer is **2**.

Problem 7. Let $T = $ TNYWR, and let $b = T^3$. Compute the value of x for which $\log_b(x+T) = \log_T(T+1)$.

Solution 7. Note that, in general, if $\log_p(q)$ is defined, then for $k > 0$, $\log_p(q) = \log_{p^k}(q^k)$. Applying this fact to the right-hand side of the given equation, it follows that $\log_T(T+1) = \log_{T^3}(T+1)^3 = \log_b(T+1)^3$. Thus the given equation implies that $x = (T+1)^3 - T$. With $T = 2$, the answer is therefore $x = 3^3 - 2 = \mathbf{25}$.

Problem 15. A (nondegenerate) triangle has sides of length 20, 26, and x, where x is a positive integer. Compute the number of possible values of x.

Solution 15. By the Triangle Inequality, it follows that $x > 26 - 20 = 6$ and $x < 26 + 20 = 46$. Thus the possible values of x are 7, 8, 9, ..., 44, 45, and this list contains $45 - 7 + 1 = \mathbf{39}$ distinct integers.

Problem 14. Let $T = $ TNYWR. Compute $\operatorname{lcm}(20, 26, T)$.

Solution 14. Note that $\operatorname{lcm}(20, 26) = 260$. If T contains a prime factor of any of 2, 5, or 13, each of these can be divided out to give remaining factors that must divide the lcm. With $T = 39 = 3 \cdot 13$, the 13 can be disregarded, hence the lcm of 20, 26, and T is $260 \cdot 3 = \mathbf{780}$.

Problem 13. Let $T = $ TNYWR. Isosceles trapezoid $ABCD$ has a perimeter of $\frac{T}{10}$, $AB = 20$, $CD = 26$, and $\overline{AB} \parallel \overline{CD}$. Let M and N be the respective midpoints of $\overline{AD}$ and $\overline{BC}$. Compute $AM + MN - NC$.

Solution 13. First, because $\overline{MN}$ is a midline, it follows that $MN = \frac{AB+CD}{2} = 23$. Let $AD = BC = 2x$. Then, because M and N are respective midpoints of $\overline{AD}$ and $\overline{BC}$, it follows that $AM = NC = x$. Thus the answer is $x + 23 - x = \mathbf{23}$ (independent of T).

Problem 12. Let $T = $ TNYWR. Let $i = \sqrt{-1}$, and for positive integers n, let $S_n = 1 + 2 + 3 + \cdots + (n-1) + n$. Compute the number of integers k such that $1 \leq k \leq 50$ and $i^T = i^{S_k}$.

Solution 12. Consider the values of $S_n = \frac{n^2+n}{2}$ modulo 4:

$$1,\ 3,\ 2,\ 2,\ 3,\ 1,\ 0,\ 0,\ 1,\ 3,\ \ldots.$$

Note that this sequence is periodic with period 8. With $T = 23$, $i^{23} = -i$, so $i^{S_k} = -i$ when $k \equiv 3 \bmod 4$, which occurs when $k \equiv 2 \bmod 8$ or $k \equiv 5 \bmod 8$. There are **13** such values of k with $1 \leq k \leq 50$ (namely 2, 5, 10, 13, 18, 21, 26, 29, 34, 37, 42, 45, 50).

Problem 11. Let $T = $ TNYWR. Compute the sum of all real number(s) x that satisfy $\sqrt{x+T+4} = x - T$.

Solution 11. Square each side of the given equation and rearrange to get $x^2 - (2T+1)x + (T^2 - T - 4) = 0$. By Vieta's Formulas, the sum of the solutions to *this* equation is $2T + 1$, but squaring the original equation may have introduced an extraneous solution. With $T = 13$, above the quadratic equation becomes $x^2 - 27x + 152 = (x-19)(x-8) = 0$. The solution $T = 19$ checks with the original equation, but $T = 8$ is extraneous, hence the answer is **19**.

Problem 10. Let $T = $ TNYWR, and let $J = 10T - 10$. In triangle LEO, $\angle E$ is a right angle, $EO = J$, and $LO = J + 1$. Compute LE.

Solution 10. Applying the Pythagorean Theorem yields $LE = \sqrt{LO^2 - EO^2} = \sqrt{(J+1)^2 - J^2} = \sqrt{2J+1}$. With $T = 19$, $J = 180$, so the answer is $\sqrt{2 \cdot 180 + 1} = \mathbf{19}$.

Problem 9. Let T = TNYWR. Speed reader Vince reads the dictionary at an impressive rate of T pages per 5 minutes. At this rate, after h hours have elapsed, where h is an integer, Vince will have first read at least 2026 pages of the dictionary. Compute h.

Solution 9. Vince reads at a rate of $\frac{60}{5} \cdot T = 12T$ pages per hour. The desired value of h is therefore $\left\lceil \frac{2026}{12T} \right\rceil$. With $T = 19$, this yields $h = \mathbf{9}$.

Problem 8. Let A be the number you will receive from position 7, and let B be the number you will receive from position 9. Let $M = \max\left\{\sqrt{A}, \sqrt{B}\right\}$, and let $m = \min\left\{\sqrt{A}, \sqrt{B}\right\}$. Two spheres are tangent and have radii M and m. Let d_{max} be the greatest possible distance between the centers of these spheres, and let d_{min} be the least possible distance between the centers of these spheres. Compute $d_{\text{max}} \cdot d_{\text{min}}$.

Solution 8. Note that $d_{\text{max}} = M + m$ when the spheres are externally tangent, and $d_{\text{min}} = M - m$ when the spheres are internally tangent. Thus $d_{\text{max}} \cdot d_{\text{min}} = M^2 - m^2$. Because $\left\{\sqrt{A}, \sqrt{B}\right\} = \{M, m\}$, it follows that $M^2 - m^2 = \left|\left(\sqrt{A}\right)^2 - \left(\sqrt{B}\right)^2\right| = |A - B|$. With $A = 25$ and $B = 9$, the final answer is $\mathbf{16}$.

15 Tiebreaker Round

Problem 1. Suppose that m and n are positive integers such that 99 is a divisor of

$$\gcd(20m + 26n + 73, 17m + 76n + 12).$$

Compute the least possible value of mn.

Problem 2. Suppose that positive real numbers a, b, and c satisfy

$$\frac{a}{b+c+4} = \frac{b}{c+a+8} = \frac{c}{a+b+12} = \frac{a+b+c}{4} - 4.$$

Compute $a + b + c$.

Problem 3. Compute the least integer n for which

$$1^2 + 2^2 - 3^2 - 4^2 + 5^2 + 6^2 - 7^2 - 8^2 + \cdots + (-1)^{\left\lfloor \frac{n-1}{2} \right\rfloor} \cdot n^2$$

is greater than 2026.

16 Tiebreaker Round Answers

Answer 1. 442

Answer 2. $3 + \sqrt{201}$

Answer 3. 46

17 Tiebreaker Round Solutions

Problem 1. Suppose that m and n are positive integers such that 99 is a divisor of

$$\gcd(20m + 26n + 73, 17m + 76n + 12).$$

Compute the least possible value of mn.

Solution 1. An equivalent condition on m and n is that they satisfy the system of congruences

$$\begin{aligned} 20m + 26n + 73 &\equiv 0 \pmod{99} \\ 17m + 76n + 12 &\equiv 0 \pmod{99}. \end{aligned}$$

Suppose that m and n satisfy these congruences. Reducing modulo 9 gives

$$\begin{aligned} 2m - n + 1 &\equiv 0 \pmod 9 \\ -m + 4n + 3 &\equiv 0 \pmod 9. \end{aligned}$$

Adding twice the second congruence to the first gives $7n + 7 \equiv 0 \pmod 9$, so $n \equiv -1 \pmod 9$. Feeding this into the second congruence above yields $m \equiv 4n + 3 \equiv -1 \pmod 9$. Let $m = 9a - 1$ and $n = 9b - 1$ for some positive integers a and b. Plugging this back in to the original system of congruences yields (after reducing coefficients modulo 99)

$$\begin{aligned} 81a + 36b + 27 &\equiv 0 \pmod{99} \\ 54a + 90b + 18 &\equiv 0 \pmod{99}, \end{aligned}$$

which is equivalent to

$$\begin{aligned} 9a + 4b + 3 &\equiv 0 \pmod{11} \\ 6a + 10b + 2 &\equiv 0 \pmod{11}. \end{aligned}$$

Multiplying the first congruence by 5 and the second congruence by 2 both yield

$$a - 2b + 4 \equiv 0 \pmod{11},$$

so the original system is equivalent to this single congruence. The solutions (a, b) modulo 11 are $(7, 0)$, $(9, 1)$, $(0, 2)$, $(2, 3)$, $(4, 4)$, $(6, 5)$, $(8, 6)$, $(10, 7)$, $(1, 8)$, $(3, 9)$, and $(5, 10)$. Except for $(7, 0)$ and $(0, 2)$, these ordered pairs are also the smallest possible pair (a, b) of positive integers in the residue classes of the solutions. For $(7, 0)$ and $(0, 2)$, the smallest pairs of positive integers in the residue classes are $(7, 11)$ and $(11, 2)$, respectively. Larger pairs in a fixed residue class will lead to larger products $(9a - 1)(9b - 1)$, so these are the only pairs to consider. It suffices to determine which of these pairs has the least possible value of $(9a - 1)(9b - 1)$. The pair $(2, 3)$ leads to a smaller product than any pair with one component at least 2 and the other component at least 3, so that leaves $(9, 1)$, $(2, 3)$, and $(1, 8)$ as candidates. Similarly, $(1, 8)$ leads to a smaller product than $(9, 1)$, so that leaves $(2, 3)$ and $(1, 8)$ as candidates. Check that $(9 \cdot 2 - 1)(9 \cdot 3 - 1) = 442$ while $(9 \cdot 1 - 1)(9 \cdot 8 - 1) = 568$, so the least possible product is **442** (attained when $m = 17$ and $n = 26$).

Problem 2. Suppose that positive real numbers a, b, and c satisfy

$$\frac{a}{b + c + 4} = \frac{b}{c + a + 8} = \frac{c}{a + b + 12} = \frac{a + b + c}{4} - 4.$$

Compute $a + b + c$.

Solution 2. Let $s = a + b + c$. The key idea is that if

$$\frac{a}{b+c+4} = \frac{b}{c+a+8} = \frac{c}{a+b+12},$$

then this quantity also equals the sum of the ratios

$$\frac{a+b+c}{(b+c+4)+(c+a+8)+(a+b+12)} = \frac{s}{2s+24}.$$

Thus

$$\frac{s}{4} - 4 = \frac{s}{2s+24} \iff s^2 - 6s - 192 = 0.$$

Solving the above quadratic equation and taking the positive solution gives $s = \mathbf{3 + \sqrt{201}}$.

Note: In fact, the exact values of a, b, and c can be calculated, and they are indeed positive. For example, as established in the solution,

$$\frac{a}{b+c+4} = \frac{a}{s-a+4} = \frac{s}{2s+24}. \qquad (*)$$

Note that $(*)$ is a linear equation in a because s is fixed. Using $s = 3+\sqrt{201}$, it follows that $a = -\frac{9}{5} + \frac{7\sqrt{201}}{15} \approx 4.8161$, and similarly, $b = 1 + \frac{\sqrt{201}}{3} \approx 5.7258$ and $c = \frac{19}{5} + \frac{\sqrt{201}}{5} \approx 6.6355$.

Problem 3. Compute the least integer n for which

$$1^2 + 2^2 - 3^2 - 4^2 + 5^2 + 6^2 - 7^2 - 8^2 + \cdots + (-1)^{\left\lfloor \frac{n-1}{2} \right\rfloor} \cdot n^2$$

is greater than 2026.

Solution 3. Let S_n denote the given sum:

$$S_n = 1^2 + 2^2 - 3^2 - 4^2 + 5^2 + 6^2 - 7^2 - 8^2 + \cdots + (-1)^{\left\lfloor \frac{n-1}{2} \right\rfloor} \cdot n^2.$$

Group every four terms, skipping the first one: $2^2-3^2-4^2+5^2 = (2+3)(2-3)+(4+5)(5-4) = -(2+3)+(4+5) = 4$. Similarly, $6^2 - 7^2 - 8^2 + 9^2 = (6+7)(6-7) + (8+9)(9-8) = 4$. Observe that this pattern continues.

Thus $S_{4k+5} = S_{4k+1} + 4$ for all nonnegative integers k and, because $S_1 = 1$, it follows that $S_{4k+1} = 4k+1$ for all nonnegative integers k. Furthermore, it follows that $S_{4k+2} = S_{4k+1} + (4k+2)^2 = 4k+1+(4k+2)^2 = (4k+2)(4k+3) - 1$.

Observe also that $S_{4k+4} < S_{4k+3} < S_{4k+2}$. Thus the least n such that $S_n > 2026$ is the least n of the form $4k+2$ such that $n(n+1) - 1 > 2026$. Because $42 \cdot 43 - 1 < 2026 < 46 \cdot 47 - 1$, the answer is **46**.

Alternate Solution: After checking the first few partial sums of the given series, observe that $S_1 = 1$, $S_5 = 5$, and $S_9 = 9$. The following argument by induction shows that for integers $n \geq 0$,

$$S_n = \sum_{k=1}^{4n+1} (-1)^{\left\lfloor \frac{k-1}{2} \right\rfloor} \cdot k^2 = 4n+1. \qquad (*)$$

The base case of $n = 0$ is readily verified. As the induction hypothesis, assume that $(*)$ holds for some nonnegative integer j. Then

$$\begin{aligned}
S_{j+1} &= \sum_{k=1}^{4j+5} (-1)^{\left\lfloor \frac{(k-1)}{2} \right\rfloor} \cdot k^2 \\
&= 4j+1+\sum_{k=4j+2}^{4j+5} (-1)^{\left\lfloor \frac{(k-1)}{2} \right\rfloor} \cdot k^2 \\
&= 4j+1+(4j+2)^2-(4j+3)^2-(4j+4)^2+(4j+5)^2 \\
&= (4j+1)+(16j^2+16j+4)+(-16j^2-24j-9)+(-16j^2-32j-16)+(16j^2+40j+25) \\
&= 4j+5 = 4(j+1)+1,
\end{aligned}$$

as desired. Thus $S_{45} = 45$, and because $44^2 = 1936$, it is not possible for the partial sums to exceed 2026 before this 45th term. Therefore the 46th partial sum must be the first to exceed 2026.